

SIMATS ENGINEERING (SAVEETHA SCHOOL OF ENGINEERING)

Department of Artificial Intelligence and Data Science

ASSESSMENT TOOL 3 – Comparative Analysis

Course Code:	DSA03	Course Title:	Natural Language Processing
CO Assessed:	CO4 – Precision	Assessment:	Assessment Tool 3 – Comparative Analysis
Weightage:	30% of CIA		
CO5 Statement:	Formulate context-free grammars, feature structures, probabilistic parsing techniques and to construct parsing frameworks. (BL-4)		
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Section / Batch:	Slot D/ 2024	Date:	27/8/26

Code of Conduct

I A Syed Thoufiq / 192472185 (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the Student: A Syed Thoufiq

Instructions

- Read each scenario carefully before answering.
- Identify the NLP techniques being compared in the question.
- Analyze each approach based on its working principles, advantages, limitations, and applicability.
- Compare the alternatives using technical reasoning rather than listing definitions.
- Support your analysis with examples from the given scenario.
- Duration: 30 minutes.

Section / Topic	Focus	Question No.
Representing Meaning & Meaning Structure of Language	Analyze sentence representations, compare structural representations of language, and evaluate how different parsing approaches capture meaning and relationships among words.	Q1
Syntax-Driven Semantic Analysis	Analyze parsing strategies for incomplete and ambiguous inputs, evaluate parsing efficiency, and determine suitable methods for real-time language understanding.	Q2
Lexemes and Their Senses & Word Sense Disambiguation	Analyze ambiguity in natural language sentences, evaluate ambiguity resolution techniques, and determine the most effective approach for selecting intended meanings.	Q3
Representing Linguistically Relevant Concepts	Analyze grammatical constraints, agreement features, argument structures, and linguistic representations required for accurate language processing.	Q4
Information Retrieval & Syntax-Driven Semantic Analysis	Analyze large-scale parsing strategies, evaluate performance trade-offs, and justify efficient approaches for processing massive linguistic datasets.	Q5

Annexure A – Comparative Analysis

1. A conversational AI system must identify grammatical relationships in user queries containing long-distance dependencies. The developers are comparing constituency parsing and dependency parsing.

Analyze how the two representations differ in capturing hierarchical and word-to-word relationships, and justify which approach is more suitable for extracting meaningful relationships from conversational text.

2. An NLP system processes partially typed search queries such as “best hotels near...” and “book a flight from Chennai to...”. The system currently uses conventional top-down parsing, while the developers are considering Earley parsing.

Analyze how the two approaches behave when input is incomplete or ambiguous, and justify which parsing strategy is more appropriate for interactive NLP applications.

3. A grammar-checking system must distinguish between multiple interpretations of structurally ambiguous sentences. The developers are comparing CFG, PCFG, and neural constituency parsers.

Analyze how each approach represents and resolves syntactic ambiguity, and justify which approach would provide the best balance between interpretability and practical accuracy.

4. A multilingual NLP system must process sentences in which grammatical information depends on number, gender, tense, and person. The developers are considering feature structures and traditional context-free grammar rules.

Analyze how feature structures extend basic grammar representations and justify their usefulness for enforcing grammatical constraints in multilingual systems.

5. A voice assistant must interpret commands containing verbs with different argument requirements, such as “give the book to John,” “sleep,” and “put the file on the desk.” The developers are considering subcategorization frames.

Analyze how subcategorization information helps determine valid sentence structures and justify its importance in semantic and syntactic analysis.

Rubrics for Calculation

Criteria	Weightage (%)	Level 4 (Exemplary)	Level 3 (Proficient)	Level 2 (Developing)	Level 1 (Beginning)
Scope of Items	20%	Selects highly relevant items with clear scope and justification	Selects appropriate items with minor gaps	Selection is limited or partially relevant	Items are unclear or not relevant
Comparison Depth	25%	Provides detailed and comprehensive comparison across multiple aspects	Provides adequate comparison with some depth	Limited comparison with few aspects	Very minimal or no comparison
Use of Evidence	20%	Supports comparison with accurate data, examples, or references	Uses some supporting evidence with minor gaps	Limited or weak supporting evidence	No supporting evidence provided
Critical Thinking	20%	Demonstrates strong critical thinking with clear interpretation and reasoning	Shows reasonable interpretation with minor gaps	Basic interpretation; lacks depth	No meaningful interpretation
Clarity of Presentation	15%	Well-organized, clear, and logically structured presentation	Generally clear with minor issues	Somewhat unclear or poorly structured	Disorganized and difficult to understand

Q. No. / Criteria Level	Scope of Items	Comparison Depth	Use of Evidenc e	Critical Thinking	Clarity of Present ation	Sub. Total	Total %
1	3	3	4	3	3	16	80
2	3	3	3	3	4	16	
3	3	3	3	4	3	16	
4	3	4	3	3	3	16	
5	4	3	3	3	3	16	

Faculty In-charge	Course Coordinator	Program Director
Dr. T Kumaragurubaran		
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Date: 27/8/26	Date: _____	Date: _____

Question 1 – Constituency Parsing vs Dependency Parsing

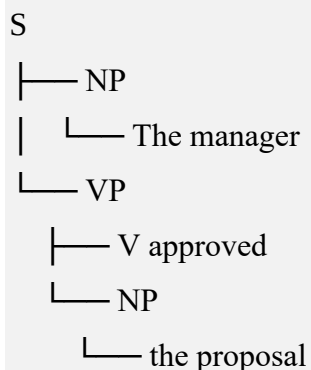
Scenario: A conversational AI system must identify grammatical relationships in user queries containing long-distance dependencies. The developers compare constituency parsing and dependency parsing.

1 Constituency Parsing

Constituency parsing represents a sentence as nested constituents such as noun phrases (NP), verb phrases (VP), prepositional phrases (PP), and clauses. The representation emphasizes hierarchical phrase structure.

Example:

"The manager approved the proposal."



Aspect	Constituency Parsing
Representation	Hierarchical phrase/constituent tree
Main unit	Phrase or constituent
Strength	Shows nested grammatical structure
Semantic use	Useful for phrase-level interpretation and clause structure
Limitation	May require additional processing to obtain direct word-to-word relations

2 Dependency Parsing

Dependency parsing represents a sentence through directed relationships between individual words. Each dependent word is linked to a head word, making grammatical relations such as subject, object and modifier explicit.

Example:

manager -> approved (subject)

proposal -> approved (object)
the -> proposal (determiner)
the -> manager (determiner)

Aspect	Dependency Parsing
Representation	Head-dependent word graph/tree
Main unit	Word-to-word dependency
Strength	Directly exposes grammatical relations
Semantic use	Useful for extracting subject, object, modifier and predicate relationships
Limitation	Phrase hierarchy is less explicit than in a constituency tree

3 Comparison

Factor	Constituency	Dependency
Hierarchical structure	Excellent	Moderate
Direct word relations	Indirect	Excellent
Phrase boundaries	Explicit	Less explicit
Subject/object extraction	Requires traversal of tree	Usually direct from dependency arcs
Long-distance relations	Can represent them through nested constituents	Can represent them as dependency links
Conversational semantic extraction	Useful for phrase interpretation	Usually more direct for relation extraction

4 Recommendation

For the stated conversational-AI objective of extracting meaningful relationships among words, dependency parsing is the more directly useful representation. It exposes head-dependent relations such as who performed an action, what was affected, and which phrase modifies another word. Constituency parsing remains valuable when the application needs detailed phrase hierarchy or clause boundaries.

A practical system can therefore use dependency parsing as the main relation-extraction representation and constituency information when phrase-level structure is important.

Question 2 – Top-Down Parsing vs Earley Parsing for Incomplete Queries

Scenario: An NLP system processes partially typed queries such as “best hotels near...” and “book a flight from Chennai to...”. It currently uses conventional top-down parsing and is considering Earley parsing.

1 Top-Down Parsing

Top-down parsing begins with the start symbol and predicts the grammar rules that could generate the input. It is straightforward when the grammar is simple and the input is complete, but ambiguity and incomplete input can cause backtracking or failed derivations.

```
S
↓
NP VP
↓
VP ...
↓
Expected continuation
```

2 Earley Parsing

Earley parsing is a chart-parsing method for general context-free grammars. It maintains partial states at input positions and supports prediction, scanning and completion. This is particularly useful when the input is incomplete or ambiguous.

Earley state form:

$A \rightarrow \alpha \cdot \beta$, origin

- before β : symbols are still expected
- completed state: β has been recognized
- chart position: where the state is currently being processed

3 Incomplete Input Analysis

Input	Top-Down Behavior	Earley Behavior
best hotels near...	May expect a complete NP/PP and fail or backtrack	Can retain partial PP and wait for continuation
book a flight from Chennai to...	May fail when destination NP is incomplete	Can preserve the state expecting a destination

4 Ambiguity Analysis

1. Top-down parsing may explore alternative productions when more than one rule can match the current phrase.
2. Repeated backtracking can increase processing time.
3. Earley parsing keeps competing partial states in a chart rather than repeatedly rebuilding the same partial derivations.
4. Earley can therefore handle general CFG ambiguity more systematically.

5 Comparison

Factor	Top-Down Parsing	Earley Parsing
Incomplete input	Less suitable	Well suited to partial states
Ambiguity	Can require backtracking	Maintains alternatives in chart
General CFG support	Depends on implementation	Designed for general CFGs
Repeated computation	Can occur during backtracking	Chart avoids much repeated work
Interactive NLP	Simple for restricted grammars	More robust for ambiguous/partial queries
Implementation complexity	Lower	Higher

6 Recommendation

Earley parsing is more appropriate for the interactive NLP application described in the scenario. The system must interpret incomplete search queries while the user is still typing. Earley's ability to retain partial parses makes it possible to maintain expected continuations such as a hotel location or flight destination.

Question 3 – CFG vs PCFG vs Neural Constituency Parsing

Scenario: A grammar-checking system must distinguish between multiple interpretations of structurally ambiguous sentences. The developers compare CFG, PCFG and neural constituency parsers.

1 CFG

A Context-Free Grammar defines which syntactic structures are valid. It uses productions such as $S \rightarrow NP VP$ and can generate multiple valid parses when a sentence is ambiguous.

CFG:

$S \rightarrow NP VP$

NP → Det N | NP PP

VP → V NP | VP PP

PP → P NP

A CFG can represent ambiguity but does not inherently rank the alternatives.

2 PCFG

A Probabilistic Context-Free Grammar assigns probabilities to grammar productions. Competing parses can therefore be ranked according to the probability of the derivation.

$P(S \rightarrow NP VP) = 0.9$

$P(VP \rightarrow V NP) = 0.7$

$P(VP \rightarrow VP PP) = 0.3$

PCFG improves ambiguity resolution when probability estimates are representative of the target language, while retaining explicit grammar rules and relatively high interpretability.

3 Neural Constituency Parsing

Neural constituency parsers learn representations from data and predict constituent structures using contextual features. They can capture complex lexical and contextual patterns that are difficult to encode manually in a grammar.

Factor	CFG	PCFG	Neural Constituency Parser
Ambiguity representation	Multiple valid parses	Multiple parses + probabilities	Predicted best structure using learned representations
Interpretability	High	High to moderate	Lower
Data requirement	Low/grammar-driven	Needs probability estimates	Usually requires substantial labeled/training data
Context sensitivity	Limited	Limited to probability model	Strong
Manual control	High	High	Lower
Practical accuracy	Depends on grammar	Better than plain CFG when probabilities fit	Often strongest with sufficient data

4 Recommended Balance

For the requested balance between interpretability and practical accuracy, PCFG is a strong middle-ground. It retains explicit grammar productions, making the selected parse easier to explain than a purely neural model, while probabilities provide a mechanism for choosing among competing structures.

If maximum practical accuracy is the primary objective and sufficient training data and computational resources are available, a neural constituency parser may be preferable. For a grammar-checking system where explanations and traceable decisions are important, PCFG offers a particularly useful compromise.

Question 4 – Feature Structures vs Traditional CFG Rules

Scenario: A multilingual NLP system processes sentences where grammatical information depends on number, gender, tense and person. The developers compare feature structures with traditional context-free grammar rules.

1 Traditional CFG Limitation

Basic CFG rules describe category combinations but do not naturally encode detailed grammatical features. For example, a rule $S \rightarrow NP VP$ can allow an NP and VP combination without explicitly requiring agreement between their person and number features.

Basic CFG:

$S \rightarrow NP VP$

$NP \rightarrow Det N$

$VP \rightarrow V NP$

2 Feature Structures

Feature structures augment grammatical categories with attribute-value information. Typical features include NUMBER, GENDER, PERSON and TENSE.

NP:

CATEGORY = NP

NUMBER = singular

PERSON = 3

VP:

CATEGORY = VP

TENSE = present

SUBJECT_FEATURES:

NUMBER = singular

PERSON = 3

3 Agreement Constraint

S -> NP VP

Constraint:

NP.NUMBER = VP.SUBJECT.NUMBER

NP.PERSON = VP.SUBJECT.PERSON

The feature values can be unified during parsing. If the values are incompatible, the parse can be rejected.

4 Multilingual Usefulness

1. Number agreement can distinguish singular and plural forms.
2. Gender features can be represented where the language marks grammatical gender.
3. Person features support first-, second- and third-person agreement.
4. Tense features allow the grammar to distinguish present, past and other temporal forms.
5. The same grammar framework can represent language-specific feature constraints without creating an entirely separate rule for every combination.
6. Feature unification can reject grammatically inconsistent structures.

5 Comparison

Factor	Traditional CFG	Feature Structures
Category representation	Nonterminals	Nonterminals + attributes
Agreement	Must be encoded indirectly	Explicitly constrained
Multilingual grammar	Can become rule-heavy	Features provide reusable constraints
Interpretability	High	High
Grammar size	Can grow with combinations	Can be more compact
Error prevention	Limited for agreement	Strong when features are correctly defined

6 Recommendation

Feature structures are more appropriate for the multilingual scenario because they explicitly represent linguistically relevant properties and enforce agreement constraints during parsing. Traditional CFG rules

remain the foundation for structural categories, while feature structures extend them with the information needed for accurate multilingual processing.

Question 5 – Sub categorization Frames for Voice Assistant Commands

Scenario: A voice assistant must interpret commands containing verbs with different argument requirements, including “give the book to John,” “sleep,” and “put the file on the desk.” The developers are considering subcategorization frames.

1 What Is a Subcategorization Frame?

A subcategorization frame specifies the types and number of arguments that a verb normally requires. It links lexical information about the verb with the syntactic structures that can realize its arguments.

Verb	Frame	Arguments
give	V NP PP	giver/agent + thing given + recipient
sleep	V	experiencer/agent only; no direct object required
put	V NP PP	agent + object + location

2 Command 1 – Give the Book to John

Verb: give

Frame: V + NP + PP

Agent = implicit speaker/user

Theme = the book

Recipient = John

The prepositional phrase “to John” is important because it identifies the recipient. A parser using the give frame expects both an object and a recipient-related phrase.

3 Command 2 – Sleep

Verb: sleep

Frame: V

No direct object is required.

A system that incorrectly requires an object for every verb would reject this valid command. The sleep frame demonstrates why verbs cannot all be assigned the same argument structure.

4 Command 3 – Put the File on the Desk

Verb: put

Frame: V + NP + PP

Agent = implicit speaker/user

Theme = the file

Location = on the desk

The location phrase is required by the verb's argument structure in this use. “Put the file” is generally incomplete because the destination/location is an essential part of the event specification.

5 Comparison

Command	Required Structure	Semantic Roles
give the book to John	V + NP + PP	Agent, Theme, Recipient
sleep	V	Agent/Experiencer
put the file on the desk	V + NP + PP	Agent, Theme, Location

7.6 Importance for Semantic Analysis

1. Prevents invalid argument structures from being accepted.
2. Helps distinguish verbs that require different numbers and types of complements.
3. Supports semantic-role assignment such as Agent, Theme, Recipient and Location.
4. Improves interpretation of voice-assistant commands.
5. Allows the system to detect missing information and ask clarification questions.
6. Supports mapping from language to structured actions.

7 Recommended Architecture

Voice Command

|

v

Speech Recognition

|

v

Verb Identification

|

v

Subcategorization Frame Lookup

|

v

Argument Detection

|

v

Semantic Role Assignment

|

v

Frame Validation

|

+---- Missing required argument ---> Clarification

|

v

Structured Action

For example, if a user says “put the file,” the system can detect that the put frame expects a location and ask, “Where should I put the file?” This is more reliable than treating all verbs as if they accept the same syntactic pattern.

Overall Comparative Analysis

Question	Alternatives	Preferred Approach	Reason
Q1	Constituency vs Dependency	Dependency for relation extraction	Direct word-to-word grammatical relations are useful for conversational meaning extraction
Q2	Top-down vs Earley	Earley	Handles incomplete and ambiguous input using chart states
Q3	CFG vs PCFG vs Neural	PCFG for interpretability/accuracy	Ranks alternatives while retaining explicit

		balance	grammar structure
Q4	CFG vs Feature Structures	Feature Structures	Enforces number, gender, tense and person constraints
Q5	Generic verb parsing vs Subcategorization	Subcategorization frames	Encodes verb-specific argument requirements and semantic roles

Final Recommendations

Application Need	Recommended NLP Component
Conversational relationship extraction	Dependency parsing
Incomplete interactive queries	Earley parsing
Interpretable ambiguity resolution	PCFG
High-accuracy parsing with sufficient training data	Neural constituency parsing
Multilingual grammatical constraints	Feature structures
Verb-specific command interpretation	Subcategorization frames

A production NLP system can combine these techniques rather than selecting only one. For example, an interactive assistant can use Earley parsing for partial syntax, feature structures for agreement, subcategorization frames for verb arguments, and dependency relations for semantic extraction.

11. Conclusion

The comparative analysis demonstrates that parsing decisions should be based on the requirements of the application rather than on definitions alone. Constituency and dependency parsing emphasize different structural relationships; Earley is better suited to incomplete and ambiguous input; PCFG provides a useful balance between explicit grammar and probabilistic ambiguity resolution; feature structures provide the grammatical constraints needed for multilingual processing; and subcategorization frames connect verbs with their valid argument structures.

The strongest engineering solution is therefore a layered parsing framework in which structural parsing, probabilistic ranking, linguistic constraints and semantic-role information complement one another.